

AutoVet

Full Production Stack

Layer	Technology	Version	Hosting / Provider	Production URL / Notes	Why we use it
Admin frontend (clinic staff)	React + Vite + Tailwind CSS (JavaScript)	React 19.2, Vite 7.3, Tailwind 3.4	Vercel	https://autovet-admin.vercel.app	Fast, modern screens for staff; Vite makes building quick and Tailwind keeps the design consistent
Client portal (pet owners)	React + Vite + Tailwind CSS (TypeScript)	React 19.2, Vite 6.4, Tailwind 3.4, TypeScript 5.6	Vercel	https://petwellness-web.vercel.app	Same tools as the admin app; TypeScript catches mistakes before they reach users
Backend API	Laravel (PHP) REST API	Laravel 12.59, PHP 8.2	Render (Docker, free plan)	https://autovet.onrender.com	Ready-made login, database and email tools, so we build features faster and more securely
Authentication	Laravel Sanctum (token-based)	4.3	Render (part of backend)	Separate logins for admin staff and pet owners	Built into Laravel; gives each user a secure login token
AI / Forecasting	Python: scikit-learn, pandas, NumPy	Python 3	Render (bundled in backend Docker image)	Inventory stock forecasting, run by Laravel as a Python script	Python is the standard for data science; predicts stock needs from past usage
Database	TiDB Serverless (MySQL-compatible)	MySQL 8 compatible	TiDB Cloud (free plan)	Local/offline copy uses MySQL 8.4 (Laragon)	Free, works like MySQL (what we use locally), so no code changes between local and online
File storage	Supabase Storage (S3-compatible)	-	Supabase (free plan)	Pet photos and uploads	Free cloud storage for uploads, so files are not lost when the server restarts
Email	Brevo (transactional email API)	-	Brevo (free plan, 300 emails/day)	Verification, reminders, invoices	Free plan with reliable delivery for verification and reminder emails
PDF generation	DomPDF (server), jsPDF + jspdf-autotable (browser)	DomPDF 3.1, jsPDF 4.2	Render / browser	Invoices and reports	Creates printable invoices and reports without extra paid services
Frontend data and forms	TanStack React Query, React Hook Form, Zod, React Router	React Query 5.99, React Router 7	-	Data caching, form validation, page routing	Keeps data fresh, checks forms before saving, and handles page navigation
Charts	Recharts	3.8	-	Admin dashboard and analytics	Easy, good-looking charts for the dashboard
Uptime monitoring	UptimeRobot	-	UptimeRobot (free)	Pings Render so the free server does not sleep	Free; keeps the server awake so the first visit is not slow
Source control	Git + GitHub	-	GitHub	https://github.com/and Reidcnny/AutoVet	Saves every change and lets the team work together safely
Deployment flow	Push to GitHub, automatic redeploy	-	Vercel (frontends), Render (backend)	Both redeploy on every push	No manual uploading; the live site updates automatically after every push

Installed but not enabled in production: Laravel Reverb and Pusher (real-time), Laravel Horizon (queues), Sentry (error tracking).